

Built-ins Objects in JavaScript - Assignment Solutions

PW EARNERS : Coding and Web Development

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LEC - 31

Section 1 - Math Object

1. Random Number

Question:

Use Math.random() to generate and display a random number.

Solution:

```
let randomNumber = Math.random();  
console.log(randomNumber);
```

2. Random Whole Number

Question:

Use Math.random() and Math.floor() to generate a random whole number between 1 and 10.

Solution:

```
let randomNumber = Math.floor(Math.random() * 10) + 1;  
console.log(randomNumber);
```

Output: Any whole number from 1 to 10

3. Round a Number

Question:

Use Math.round() to round the number 4.6 to the nearest integer.

Solution:

```
let number = 4.6;  
console.log(Math.round(number));
```

Output:

5

4. Floor and Ceil

Question:

Use `Math.floor()` and `Math.ceil()` on the number 7.3 and display both results.

Solution:

```
let number = 7.3;  
console.log("Floor:", Math.floor(number));  
console.log("Ceil:", Math.ceil(number));
```

Output:

Floor: 7
Ceil: 8

5. Absolute Value

Question:

Use `Math.abs()` to find the positive value of -25.

Solution:

```
let number = -25;  
console.log(Math.abs(number));
```

Output: 25

6. Power and Square Root

Question:

Use `Math.pow()` to calculate 2 raised to the power 3 and `Math.sqrt()` to find the square root of 64.

Solution:

```
console.log("Power:", Math.pow(2, 3));  
console.log("Square Root:", Math.sqrt(64));
```

Output:

Power: 8
Square Root: 8

7. Minimum and Maximum

Question:

Use `Math.min()` and `Math.max()` to find the smallest and largest values from 10, 25, 5, and 18.

Solution:

```
let numbers = [10, 25, 5, 18];  
console.log("Minimum:", Math.min(...numbers));  
console.log("Maximum:", Math.max(...numbers));
```

Output:

Minimum: 5

Maximum: 25

Section 2 - String Built-in Methods

8. Extract Part of a String

Question:

Create a string "JavaScript Programming" and use `slice()` to extract the word "JavaScript".

Solution:

```
let text = "JavaScript Programming";  
let result = text.slice(0, 10);  
console.log(result);
```

Output:

JavaScript

9. Split a String

Question:

Create a string "HTML,CSS,JavaScript" and use `split()` to separate the values.

Solution:

```
let text = "HTML,CSS,JavaScript";  
let result = text.split(",");  
console.log(result);
```

Output:

```
["HTML", "CSS", "JavaScript"]
```

10. Replace Text

Question:

Create a string "Hello World" and use `replace()` to replace "World" with "JavaScript".

Solution:

```
let text = "Hello World";  
let result = text.replace("World", "JavaScript");  
console.log(result);
```

Output:

```
Hello JavaScript
```

11. Check Email

Question:

Create a variable email containing an email address and use `includes()` to check whether it contains the `@` symbol.

Solution:

```
let email = "user@example.com";  
console.log(email.includes("@"));
```

Output:

```
true
```

12. Check File Extension

Question:

Create a variable `fileName` containing "assignment.pdf" and use `endsWith()` to check whether the file has a .pdf extension.

Solution:

```
let fileName = "assignment.pdf";  
console.log(fileName.endsWith(".pdf"));
```

Output:

true

13. Remove Extra Spaces

Question:

Create a string with extra spaces, such as " Hello JavaScript ", and use trim() to remove the spaces from the beginning and end.

Solution:

```
let text = " Hello JavaScript ";  
let result = text.trim();  
console.log(result);
```

Output:

Hello JavaScript

14. Replace a Greeting

Question:

Create a variable greet containing "Hello User" and use replace() to change "User" to a name of your choice.

Solution:

```
let greet = "Hello User";  
let result = greet.replace("User", "Rahul");  
console.log(result);
```

Output:

Hello Rahul

Section 3 - Number Built-in Methods

15. Format a Decimal Number

Question:

Create a variable containing the number 12.56789 and use toFixed(2) to display the number with two decimal places.

Solution:

```
let number = 12.56789;  
console.log(number.toFixed(2));
```

Output:

12.57

16. Format a Price

Question:

Create a variable price containing a decimal value and use toFixed(2) to display it as a price with two decimal places.

Example Input: 99.5

Output: 99.50

Solution:

```
let price = 99.5;  
console.log(price.toFixed(2));
```

Output:

99.50

Section 4 - Date Object

17. Current Date and Time

Question:

Create a Date object using new Date() and display the current date and time.

Solution:

```
let currentDate = new Date();  
console.log(currentDate);
```

Output:

The current date and time will be displayed.

18. Store a Specific Date

Question:

Create a Date object for a specific date of your choice and display it.

Example: `new Date("2026-01-01")`

Solution:

```
let specificDate = new Date("2026-01-01");  
console.log(specificDate);
```

Output:

The stored date will be displayed.

19. Current Timestamp

Question:

Use `Date.now()` to get and display the current timestamp.

Solution:

```
let timestamp = Date.now();  
console.log(timestamp);
```

Output:

A number representing the current timestamp in milliseconds will be displayed.

20. Simple Date Difference

Question:

Create two Date objects for two different dates and find the difference between them in milliseconds.

Example:

Date 1: January 1, 2026

Date 2: January 2, 2026

Solution:

```
let date1 = new Date("2026-01-01");
```

```
let date2 = new Date("2026-01-02");  
let difference = date2 - date1;  
console.log("Difference in milliseconds:", difference);
```

Output:

Difference in milliseconds: 86400000

Note: 86400000 milliseconds = 1 day.